

Multi-copy repeats: where the two-copy theory of assembly ambiguity stops

Ruqing Chen

GUT Geoservice Inc., Rivière-Beaudette, Québec J0P 1R0, Canada
 ruqing@hotmail.com

25 September 2026

Abstract

The sandpile group of a genome's de Bruijn graph splits into a length-driven part and the group $\mathcal{K}(D_w)$ of the transition digraph of the circular word of repeat occurrences [4]. For repeats of multiplicity two, $\mathcal{K}(D_w)$ is the critical group of a one-vertex map: $\text{coker}(A + I)$ for Bouchet's signed interlace matrix, by a theorem of Merino, Moffatt and Noble — a presentation whose entries are *pairwise* data about the repeats. This note asks what survives for repeats with three or more copies. Two families of any multiplicities r, s give $\mathcal{K}(D_w) = \mathbb{Z}/c$, where c is the number of transitions from one family to the other, $1 \leq c \leq \min(r, s)$; this contains the interleaving theorem for two-copy repeats. Beyond two families nothing pairwise survives: for three-copy repeats, two words with identical pairwise occurrence patterns already have groups $\mathbb{Z}/3$ and $\mathbb{Z}/4$ at $m = 3$, so no matrix with pairwise entries can present $\mathcal{K}(D_w)$; detaching a three-copy vertex into two-copy vertices does not control the group; and the literature has no interlacement formula for r -in r -out digraphs with $r \geq 3$. A census of the groups realised by three- and four-copy arrangements is given (9, 17 and 28 groups for $(3, 3, 3)$, $(4, 4, 4)$ and $(3, 3, 3, 3)$, up to $(\mathbb{Z}/3)^3$). On φ X174 the skeleton is 2-in 2-out at $k = 11, 10$, where the signed interlace matrix reproduces the group exactly, and acquires vertices of out-degree 3 to 6 from $k = 9$ down: the two-copy presentation stops exactly where the arrangement group becomes large. Four checks, all exact.

1 Introduction

Let S be a circular sequence with repeat families $W_1, \dots, W_m$ of multiplicities r_i , clean in the sense of [2], and let w be the circular word of their occurrences. The bundle-chain splitting theorem [4] gives

$$\mathcal{K}(G_k(S)) \cong \bigoplus_i (\mathbb{Z}/r_i)^{s_i-1} \oplus \mathcal{K}(D_w),$$

where D_w has one vertex per family and one arc per adjacency of occurrences. The first summand is what longer k -mers remove. The second, the *arrangement group*, is what no k removes, and for $r_i = 2$ it has a striking description: D_w is the directed medial graph of a bouquet whose edges are the repeats, and by [9, Thm. 7.3] $\mathcal{K}(D_w) \cong \text{coker}(A + I)$ with A the skew-symmetric $\{0, \pm 1\}$ matrix recording, for each pair of repeats, whether and in which order their occurrences interlace. The entries of A are pairwise data; its determinant is the interlace polynomial at 1 [1], the number of Euler circuits of D_w [8, 6].

Real genomes have repeats with more than two copies. In φ X174 the skeleton of [4] is 2-in 2-out at $k = 11$ and $k = 10$ but has vertices of out-degree 3 and 4 at $k = 9$ and up to 6 at $k = 7$ (Table 2). This note asks whether the two-copy description extends. The answer is: only in the smallest case.

What is known. For 2-in 2-out digraphs the determinant formula $\# \text{Euler circuits} = \det(I + A)$ is due to Macris and Pulé [8], with elementary proofs by Lauri and Jonsson [6]; the group statement is [9]. Jonsson [6, p. 192] refers to an unpublished report generalising the intersection matrix to arbitrary Eulerian digraphs, where the determinant counts arborescences rather than circuits; we could not locate it. Lauri’s own attempt at extensions [7] is described by its reviewer as inconclusive. For rotor-routing the picture is different: Tóthmérész [10] constructs a canonical simply transitive action of the sandpile group on compatible Euler tours of any Eulerian ribbon digraph, whatever the degrees, so the torsor of [3] generalises even though, as we show, the matrix presentation does not.

2 Setting

D_w is the transition digraph of the circular word w : vertices the letters, one arc $e \rightarrow f$ per occurrence of e immediately followed by f around the circle; loops are deleted (they do not affect the Laplacian). It is balanced with $d^\pm(e) = r_e$. Its sandpile group is $\mathcal{K}(D_w) = \text{coker}(\Delta_s^T)$ with $\Delta = D^+ - T$, T_{ef} the number of transitions $e \rightarrow f$; the isomorphism class does not depend on the sink s [5, Lem. 4.12]. Groups are compared by elementary divisors. The *pairwise pattern* of two letters e, f is the circular binary word obtained by deleting all other letters from w , taken up to rotation.

3 Two families

Theorem 1 (Two families). *Let w be a circular word over $\{A, B\}$ with r copies of A and s of B , and let c be the number of transitions $A \rightarrow B$ (equivalently the number of maximal blocks of A ’s). Then $1 \leq c \leq \min(r, s)$, every such c occurs, and*

$$\mathcal{K}(D_w) \cong \mathbb{Z}/c.$$

Proof. D_w has two vertices; the number of arcs $A \rightarrow B$ equals the number of arcs $B \rightarrow A$ (each block of A ’s is entered once and left once), namely c , and the remaining arcs are loops. Reduced at B , the Laplacian is the 1×1 matrix (c) . The word $A^{r-c+1}B^{s-c+1}(AB)^{c-1}$ realises c , and $c \leq \min(r, s)$ since each block contains at least one letter. $\square$

For $r = s = 2$ this is the interleaving theorem [2, Thm. 6]: $c = 1$ for $AABB$ and $ABBA$ (disjoint or nested, no arrangement ambiguity) and $c = 2$ for $ABAB$. For two families the arrangement ambiguity is thus a single cyclic factor whose order is a crossing number, and no sign is needed. Check (A) verifies the theorem over all 119 circular words with $r, s \leq 5$.

4 Beyond two families nothing pairwise survives

Proposition 2 (Pairwise patterns do not determine the group). *The circular words*

$$w_1 = AABACCBBC, \quad w_2 = AABABCBCC,$$

each letter three times, have the same pairwise pattern for every pair of letters, and

$$\mathcal{K}(D_{w_1}) = \mathbb{Z}/3, \quad \mathcal{K}(D_{w_2}) = \mathbb{Z}/4.$$

Consequently no square matrix whose (e, f) entry is a function of the pairwise pattern of e and f can present $\mathcal{K}(D_w)$ for three-copy repeats, even for $m = 3$.

Proof. Restricting to $\{A, B\}$, $\{A, C\}$ and $\{B, C\}$ gives, for both words, the cyclic patterns 001101, 000111 and 001101 (with the first letter of the pair written as 1). In w_1 all six off-diagonal transition counts are 1, so the Laplacian reduced at C is $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, of determinant 3 and content 1: $\mathbb{Z}/3$. In w_2 the transitions are $A \rightarrow B$ twice, $A \rightarrow C$ never, $B \rightarrow A$ once, $B \rightarrow C$ twice, giving $\begin{pmatrix} 2 & -2 \\ -1 & 3 \end{pmatrix}$, of determinant 4 and content 1: $\mathbb{Z}/4$. $\square$

Remark 3 (Why this is sharper than the two-copy failure). For two-copy repeats the pairwise pattern of e, f is a single bit (interlaced or not), and the group is likewise not a function of these bits alone — but the first failure is at $m = 6$ [9, Rem. 6.7], and it is repaired by a sign: reading from a base point, record whether the occurrences run $e f e f$ or $f e f e$. The sign depends on the base point, and the cokernel does not, because changing base point acts by pivots [9, Lem. 4.8]. For three or more copies the analogous base-pointed pairwise data — the linear restriction of w to each pair — determine the whole word for every m (they fix the relative order of any two occurrences), so “the group is a function of them” is vacuous, and the only genuine pairwise invariants are the rotation classes of Proposition 2, which fail already at $m = 3$. There is no room between the two for a sign.

Proposition 4 (No reduction to two copies). *Let G be a balanced digraph with a vertex v of in- and out-degree 3, and let G' be obtained by detaching v into v' (two of the in-arcs, one of the out-arcs) and v'' (the remaining in-arc, the other two out-arcs) joined by a new arc $v' \rightarrow v''$, so that G' is 2-in 2-out at both. Then $\mathcal{K}(G)$ is in general neither a quotient nor a subgroup of $\mathcal{K}(G')$: there is a G with $\mathcal{K}(G) = \mathbb{Z}/4 \oplus \mathbb{Z}/5$ whose nine detachments have groups $\mathbb{Z}/23$, $\mathbb{Z}/25$, $\mathbb{Z}/31$, $\mathbb{Z}/32$, $\mathbb{Z}/2 \oplus \mathbb{Z}/13$, $\mathbb{Z}/3 \oplus \mathbb{Z}/8$, $\mathbb{Z}/2 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/4$ and $\mathbb{Z}/2 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/5$.*

Over 150 random balanced digraphs and all nine detachments each, $\mathcal{K}(G)$ is a quotient of $\mathcal{K}(G')$ in 140 of 1350 cases (check (C)). Euler circuits of G' are those of G compatible with the chosen transition at v , a subset of no fixed proportion, and the group follows the count. So the arrangement group of a multi-copy skeleton cannot be computed by splitting it into a two-copy one and applying [9].

Remark 5 (What does determine it). Trivially, the transition matrix T : the arrangement group is the sandpile group of the transition digraph, which is what [4] says. The content of the two-copy theory is that T can be traded for interlacement — a different, topological parametrisation of the same group. Propositions 2 and 4 say that for three or more copies no such trade is available through pairwise data or through two-copy surrogates. A Gram-matrix route (the form $x^T(I - S)y$ of the position cycle restricted to consecutive-occurrence differences) was also tried and is not the Merino–Moffatt–Noble matrix even for two copies; it is recorded in the ancillary README so that it is not retried.

5 Census

Table 1 lists, for each multiplicity vector, the number of circular words (one per rotation class), the number of distinct transition digraphs up to relabelling, the number of distinct groups, the

largest order, and the non-cyclic groups that occur. The last column counts the classes of pairwise patterns and, in parentheses, how many of them carry more than one group. For two families the groups are exactly $\mathbb{Z}/c$, $c \leq \min(r, s)$, as Theorem 1 predicts, and every pairwise class is pure. From three families on the arrangement group is far richer than for two-copy repeats — three two-copy repeats give only $0, \mathbb{Z}/2, \mathbb{Z}/3, (\mathbb{Z}/2)^2$ [2, Prop. 8], three three-copy repeats give nine groups up to order 9 — and most pairwise classes are mixed.

mult.	words	D_w	groups	$\max \mathcal{K} $	non-cyclic groups	pairwise (mixed)
(3, 3)	4	3	3	3	—	3 (0)
(4, 4)	10	4	4	4	—	7 (0)
(3, 3, 3)	188	12	9	9	$(\mathbb{Z}/2)^2, (\mathbb{Z}/3)^2$	15 (6)
(3, 3, 2)	70	15	7	6	$(\mathbb{Z}/2)^2$	10 (4)
(2, 2, 3)	30	10	5	4	$(\mathbb{Z}/2)^2$	6 (2)
(4, 4, 4)	2896	26	17	16	$(\mathbb{Z}/2)^2, (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/3, (\mathbb{Z}/3)^2, (\mathbb{Z}/4)^2, \mathbb{Z}/2 \oplus \mathbb{Z}/4$	174 (86)
(3, 3, 3, 3)	30804	96	28	27	those, and $(\mathbb{Z}/2)^3, (\mathbb{Z}/3)^3, \mathbb{Z}/2 \oplus (\mathbb{Z}/3)^2$	180 (132)

Table 1: Arrangement groups of multi-copy repeats, exhaustive over circular words (check (B)). D_w : distinct transition digraphs. All entries exact.

6 φ X174

Table 2 gives the out-degree profile of the bundle-contracted skeleton of φ X174 (NC_001422.1) at $k = 12, \dots, 7$. At $k = 11$ and $k = 10$ every skeleton vertex has in- and out-degree 2, and the signed interlace matrix of an Euler circuit found by Hierholzer’s algorithm gives $\text{coker}(A+I) = \mathbb{Z}/3 \oplus \mathbb{Z}/11$ and $\mathbb{Z}/4 \oplus \mathbb{Z}/1076338579$, the groups computed in [4] from the Laplacian (check (D)) — the theorem of [9] on a real genome. From $k = 9$ down the skeleton has vertices of out-degree 3 (5 of them at $k = 9$, 42 at $k = 8$, 147 at $k = 7$), 4, 5 and 6: the two-copy presentation stops applying at the same k at which the skeleton grows from 37 to 110 vertices and the group acquires its large prime factors.

k	skeleton	arcs	out-degree multiset
12	1	0	(single vertex)
11	8	16	2^8
10	37	74	2^{37}
9	110	227	$2^{104} 3^5 4^1$
8	312	679	$2^{264} 3^{42} 4^5 5^1$
7	826	1945	$2^{619} 3^{147} 4^{41} 5^{12} 6^7$

Table 2: φ X174: size and out-degree profile of the skeleton ($d^n = n$ vertices of out-degree d). At $k = 11, 10$ the skeleton is 2-in 2-out and $\text{coker}(A+I)$ equals $\mathcal{K}(\text{Sk})$; below, multi-copy vertices appear. Exact.

7 Verification

The script `multicopy.py` runs the four checks of Table 3 in about forty seconds, all passing; its output is archived as `multicopy_output.txt`. All arithmetic is exact; groups are compared by

elementary divisors.

key	check	mode
(A)	Theorem 1 on all 119 two-letter circular words, $r, s \leq 5$	exact
(B)	Table 1; the witness of Proposition 2	exact
(C)	Proposition 4: 150 digraphs, 1350 detachments; the witness	exact
(D)	Table 2; $\text{coker}(A + I) = \mathcal{K}(\text{Sk})$ at $k = 11, 10$	exact

Table 3: The verification battery, 4/4.

8 Scope

Proved. Theorem 1; Propositions 2 and 4 by explicit witness.

Quoted. The splitting theorem [4]; the two-copy presentation [9]; the determinant formula [8, 6] and interlace polynomial [1]; sink-independence [5]; the rotor torsor for Eulerian ribbon digraphs [10].

Open. (a) Whether any presentation of $\mathcal{K}(D_w)$ for $r \geq 3$ exists whose entries are “local” in a weaker sense than pairwise — e.g. functions of the patterns of triples — and whether Jonsson’s unpublished generalised intersection matrix, which counts arborescences, has a cokernel equal to $\mathcal{K}$. Theorem 1 shows that at least for two families the group is a crossing number, so a multi-chord crossing calculus is not ruled out; Proposition 2 shows it cannot be pairwise. (b) The census suggests questions we have not pursued: which groups occur for (r, r, r) as r grows, and whether the largest order is always attained by an elementary abelian group. (c) The φX174 skeleton at $k = 8$ (312 vertices) is beyond the dense exact Smith form used here; a modular Smith form would extend Table 2 of [4].

Provenance

Unrefereed preprint. Theorem 1 is elementary and, as far as we can find, unstated in this form; the propositions are negative results established by explicit computation and are the reason this note exists: the natural extension of [9] to multi-copy repeats was the obvious next step after [4], and it does not exist in any pairwise form. We searched for an interlacement or determinant formula for Euler circuits of r -in r -out digraphs with $r \geq 3$ and found none; a search cannot establish a negative.

References

- [1] R. Arratia, B. Bollobás and G. B. Sorkin, *The interlace polynomial of a graph*, J. Combin. Theory Ser. B **92** (2004), 199–233.
- [2] R. Chen, *Assembly ambiguity is not a function of branch statistics: the sandpile group of a de Bruijn graph*, companion preprint (Bio 1, version 2), 2026.
- [3] R. Chen, *The sandpile group as a coordinate system on genome reconstructions*, companion preprint (Bio 2), 2026.

- [4] R. Chen, *The sandpile group of a de Bruijn graph splits along its repeats*, companion preprint (Bio 3), 2026.
- [5] A. E. Holroyd, L. Levine, K. Mészáros, Y. Peres, J. Propp and D. B. Wilson, *Chip-firing and rotor-routing on directed graphs*, in: In and Out of Equilibrium 2, Progr. Probab. **60**, Birkhäuser, 2008, 331–364.
- [6] J. Jonsson, *On the number of Euler trails in directed graphs*, Math. Scand. **90** (2002), 191–214.
- [7] J. Lauri, *On a determinant formula for enumerating Euler trails in a class of digraphs*, Rend. Sem. Mat. Messina Ser. II **5** (1999), 75–90.
- [8] N. Macris and J. V. Pulé, *An alternative formula for the number of Euler trails for a class of digraphs*, Discrete Math. **154** (1996), 301–305.
- [9] C. Merino, I. Moffatt and S. Noble, *The critical group of a combinatorial map*, Combinatorial Theory **5** (2025), no. 3; arXiv:2308.13342.
- [10] L. Tóthmérész, *On canonical sandpile actions of embedded graphs*, Advances in Combinatorics (2026), no. 6; arXiv:2504.05275.